

Bibliometric Analysis of Digital Finance and Machine Learning Research

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DIGITAL

Agenda

- Research Questions
- Related Work
- Methodology
- Results
- Conclusion



Research Questions

- 1) How has research activity at the intersection of digital finance and machine learning evolved over time?
- 2) How has this research diffused geographically across countries?



Related Work

Digital Finance and FinTech

- Conceptual foundations of digital finance and FinTech (Gomber et al., 2017)
- Bibliometric mappings of FinTech research (Salem & Shahimi, 2025; Jafri et al., 2025; Brika, 2022; Li & Xu, 2022)
- Domain-specific analyses (e.g., e-payments, technology convergence) (Bhatt et al., 2023)

AI & Machine Learning in Finance

- Bibliometric analyses of AI-in-finance and ML applications (Roy et al., 2025; Zakaria et al., 2023; Mukthar et al., 2025)

Gap: No existing study explores the development and geographical diffusion of research at the intersection of digital finance and machine learning.



Methodology

1. Defined Search Query
2. Exported bibliometric metadata from Scopus
3. Data preprocessing
4. Data visualization



Results - Trend

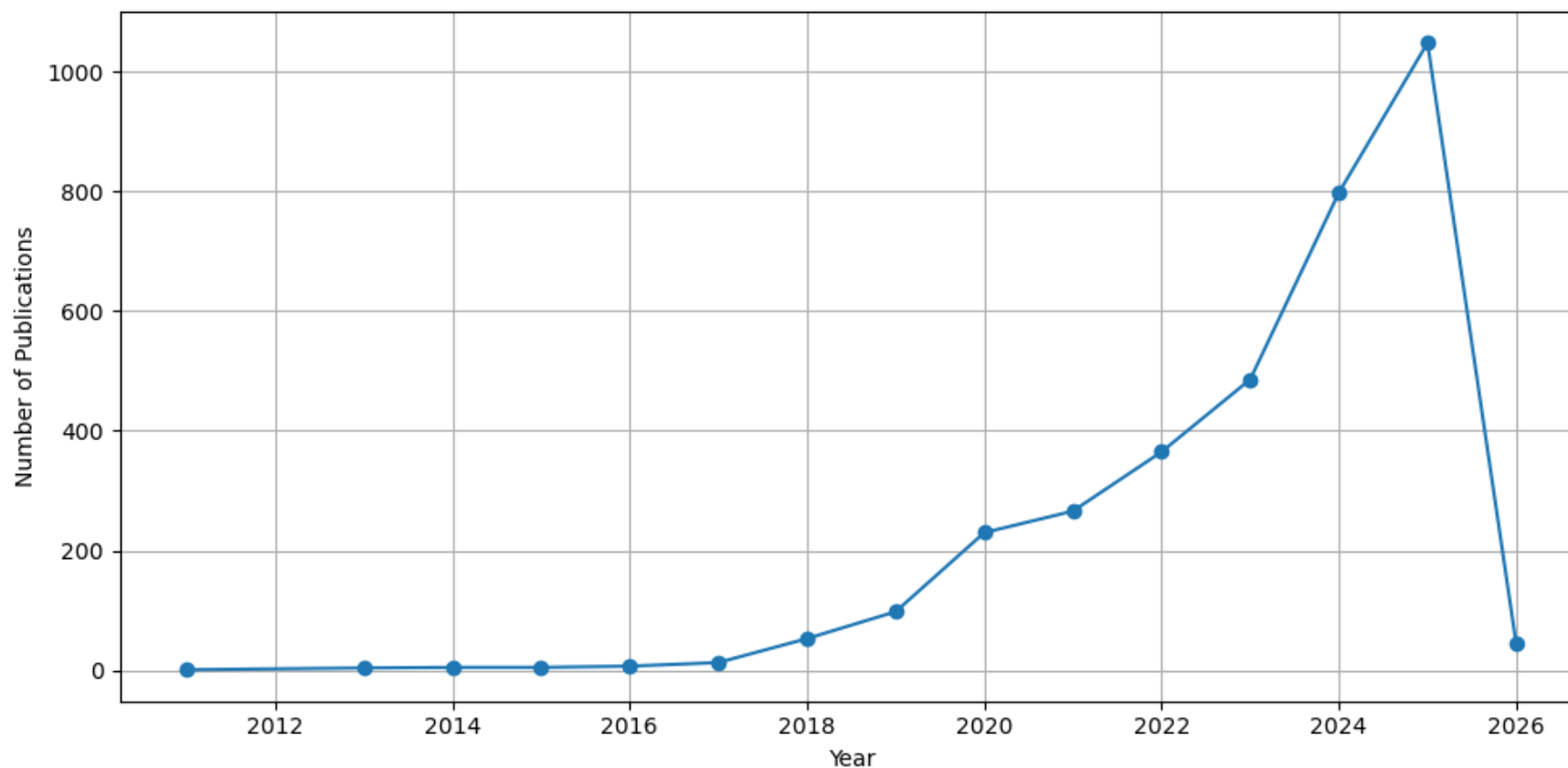


Fig 1: Annual number of publications at the intersection of digital finance and machine learning (2011--2026). The figure shows a period of modest growth during the early 2010s, followed by a sharp acceleration beginning around 2018.

Results – Country Diffusion

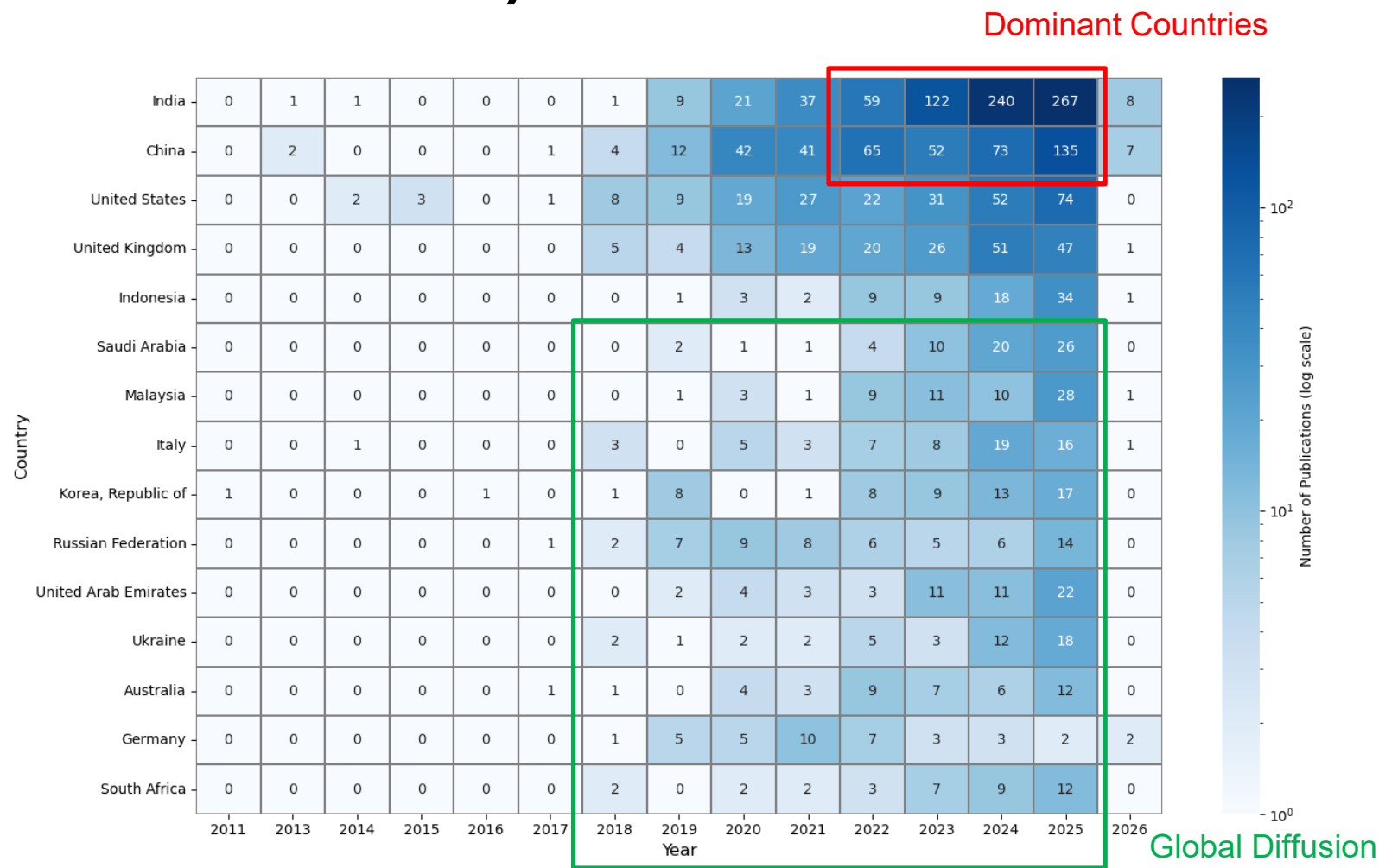


Fig 2: Heatmap of yearly publication counts for the top 15 countries in the dataset (log scale). The figure visualizes how research activity at the intersection of digital finance and machine learning has diffused geographically between 2011 and 2026.

Results – Key Word Trend

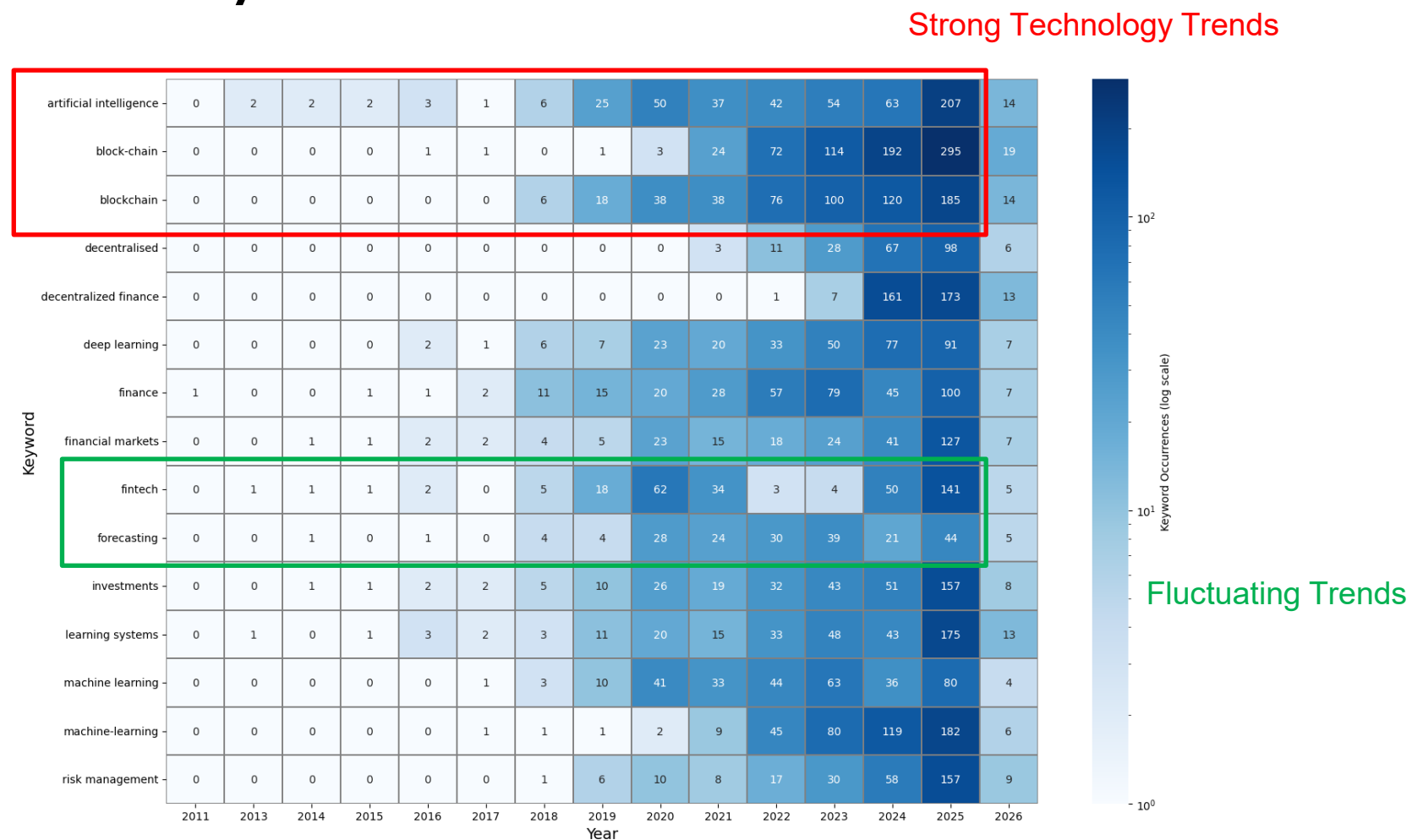


Fig 3: Heatmap of yearly occurrences for the top 15 keywords in the dataset (log scale). The figure illustrates the temporal evolution of core themes in research at the intersection of digital finance and machine learning between 2011 and 2026.

Conclusion

- Number of annual publications accelerated after 2018
- Research diffuses globally with India and China leading in terms of number of annual publications
- Keyword usage seems influenced by technology trends



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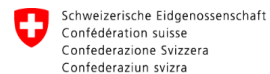


Thank you for your attention! Questions?

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